

Problem Set 1

Industrial Organization

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I. Conduct analysis in the product space: a monopsonist firm?

Part 1: theory and DGP

i. Solving for the equilibria

In scenario 1, the firm solves

$$\max_{L_t} p_t e^{\nu_t} L_t^\beta - w_t L_t,$$

with FOC

$$w_t = \beta p_t e^{\nu_t} L_t^{\beta-1},$$

which is the labor demand equation in this case. In scenario 2, where the firm knows it will affect wages, we also have to incorporate the $\partial w_t / \partial L_t$ term from the labor supply. The FOC in scenario 2 is

$$\beta p_t e^{\nu_t} L_t^{\beta-1} = w_t + L_t \frac{\partial w_t}{\partial L_t}.$$

We can already see that wages will be lower in this case. Because labor supply is upward-sloping in wages, the equation above will yield the wages of scenario 1 minus a positive term. Taking logs in the labor supply equation, we get

$$\log(L_t) = \alpha + \gamma \log(w_t) + \mu \log(u_t) + \rho \log(u_t) \log(w_t) + \eta_t.$$

This makes it easy to calculate the labor supply elasticity

$$\frac{\partial \log(L_t)}{\partial \log(w_t)} = \frac{w_t}{L_t} \frac{\partial L_t}{\partial w_t} = \gamma + \rho \log(u_t).$$

By the inverse function theorem,

$$\frac{\partial \log(w_t)}{\partial \log(L_t)} = \frac{L_t}{w_t} \frac{\partial w_t}{\partial L_t} = \frac{1}{\gamma + \rho \log(u_t)}.$$

Dividing both sides of the scenario 2 labor demand equation by w_t ,

$$\frac{\beta p_t e^{\nu_t} L_t^{\beta-1}}{w_t} = 1 + \frac{L_t}{w_t} \frac{\partial w_t}{\partial L_t} = 1 + \frac{1}{\gamma + \rho \log(u_t)}.$$

Solving for w_t ,

$$w_t = \beta p_t e^{\nu_t} L_t^{\beta-1} \quad (\text{Scenario 1 Labor Demand})$$

$$w_t = \left(1 + \frac{1}{\gamma + \rho \log(u_t)}\right)^{-1} \beta p_t e^{\nu_t} L_t^{\beta-1}. \quad (\text{Scenario 2 Labor Demand})$$

We can combine both by defining the markdown

$$M = \begin{cases} 1, & \text{in Scenario 1} \\ \left(1 + \frac{1}{\gamma + \rho \log(u_t)}\right)^{-1}, & \text{in Scenario 2} \end{cases}.$$

Then, in both cases, labor market equilibrium (L_t, w_t) is determined by both labor supply and demand, so by the system

$$\begin{cases} w_t = \beta M p_t e^{\nu_t} L_t^{\beta-1} \\ L_t = \exp\left(\alpha + \gamma \log(w_t) + \mu \log(u_t) + \rho \log(u_t) \log(w_t) + \eta_t\right) \end{cases}.$$

If we knew all the parameters $(\beta, \alpha, \gamma, \mu, \rho)$, and we observed $(p_t, u_t, \nu_t, \eta_t)$, we would also know M . Hence, the only unknowns in the two equations would be (L_t, w_t) . We could solve for them by hand by substituting w_t from the labor demand into the labor supply equation, from which we could find a root L_t^* . Then, we just plug it into either equation to compute w_t^* .

1. Wages in Scenario 1 vs. in Scenario 2

Because the markdown term M lies in $(0, 1)$, the labor demand curve is shifted down in Scenario 2. Therefore, equilibrium wages will be lower in Scenario 2.

2. Markdown

By the markdown formula

$$M = \left(1 + \frac{1}{\gamma + \rho \log(u_t)}\right)^{-1},$$

M is increasing in γ , in ρ , and does not depend on μ . It is also increasing in u_t . Because M is a multiplier, M becoming larger means that wages are being marked down less, hence higher. Here, we rely on $u_t > 1$ to avoid the logs being negative, so unemployment u_t would be defined in absolute terms and not as a fraction.

II. Demand systems for bikes?

Part 1: theory and DGP

i. Derive expression for share of consumers

Each consumer i , when consuming bike j , obtains utility

$$U_{ij} = \exp(\beta_w w_j + \beta_g g_j) - \alpha_i p_j,$$

where $\alpha_i \stackrel{i.i.d.}{\sim} \text{LogNormal}(\mu, 1)$

The probability that consumer i chooses bike j is typically

$$P_{ij} = P\{U_{ij} \geq U_{ik} \quad \forall k\}.$$

But, because there is no consumer-good idiosyncratic term, there can be ties with non-zero probability. This is very similar to the Bresnahan vertical differentiation model, but with heterogeneity in α_i that multiplies prices, instead of the v_i that multiplies qualities. To deal with this, we will reduce the good characteristics into a single measure of quality, which is all that matters for decision-making. Define the quality of good j as

$$q_j = \exp(\beta_w w_j + \beta_g g_j).$$

Then the utility obtained when consuming good j is $U_{ij} = q_j - \alpha_i p_j$. The consumer chooses from a menu of (q_j, p_j) to maximize utility, depending on α_i . Without loss of generality, we can reorder goods using $q_1 \leq q_2 \leq \dots \leq q_J$, with the possibility of ties.

If there is a tie between goods j and $j+1$, where $q_j = q_{j+1}$, one of two things will occur. The first case is $p_j = p_{j+1}$, and both are equally good choices for the consumer. In this case, we assume the consumer will flip a coin to pick each with 50% probability. If there is a perfect tie between k goods, the consumer may pick each with probability $1/k$. The second case is that $p_j \neq p_{j+1}$, so we can assume $p_j < p_{j+1}$ WLOG. But then, $U_{i,j} > U_{i,j+1}$ for all consumers i , meaning that good $j+1$ is strictly dominated. As no one will choose it, it will have zero market share. Our approach is that, in case of a perfect tie, we remove all but one of the options from the choice set. The remaining good is representative of all duplicates combined.

In general, any good k is strictly dominated by j if $(q_k, -p_k) \leq (q_j, p_j)$, element-by-element, and they are not exactly equal. We can safely remove strictly dominated goods from our menu. Since we are removing ties, this already implies that both qualities and prices much go in the same direction. We can arrange the remaining goods such that $q_1 < q_2 < \dots$ and $p_1 < p_2 < \dots$. Even more strongly, assume three consecutive goods j , $j+1$, and $j+2$, have

$$\frac{q_{j+1} - q_j}{p_{j+1} - p_j} \leq \frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}}.$$

We will show that almost no consumer will pick good $j+1$. Assume consumer i chooses $j+1$. Then, he prefers it to both j and $j+2$, meaning $q_{j+1} - \alpha_i p_{j+1} \geq q_j - \alpha_i p_j$. Rearranging,

$$\alpha_i \leq \frac{q_{j+1} - q_j}{p_{j+1} - p_j}.$$

Doing the same for $j + 1$ and $j + 2$, as $q_{j+1} - \alpha_i p_{j+1} \geq q_{j+2} - \alpha_i p_{j+2}$, then

$$\alpha_i \geq \frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}}.$$

Combining the two, this implies

$$\frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}} \leq \alpha_i \leq \frac{q_{j+1} - q_j}{p_{j+1} - p_j}.$$

Along with our initial assumption that the ratios are actually increasing, the only way both can be true is if α is exactly equal to both ratios. Because this happens with zero probability, removing goods like $j + 1$ also will not affect choice probabilities (almost everywhere), and will therefore not change market shares. This means that there is no loss to imposing the assumption

$$\frac{q_{j+1} - q_j}{p_{j+1} - p_j} > \frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}}.$$

Therefore, we construct a reduced menu $\widetilde{C}$ of available pairs $(q_j, -p_j)$. Because we are only looking at a set of $(q, -p)$ choices and not individual goods, there are no perfect ties by construction. We also rule out strict dominance by removing goods j such that $(q_j, -p_j) \leq (q_k, -p_k)$ for some $k \neq j$. Goods are then placed in a strictly increasing order of both quality and prices. Finally, we remove the possibility of increasing slopes by, given this ordering, removing any good $j + 1$ such that

$$\frac{q_{j+1} - q_j}{p_{j+1} - p_j} \leq \frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}}.$$

This is done iteratively: we order the goods and compute the slopes. Then, we remove any non-increasing slopes, reassign the order and repeat. When choosing from this menu, consumers will have a unique best choice with probability 1. To see why, assume a consumer is indifferent between $(q, -p)$ and $(q', -p')$. Then, $q - \alpha_i p = q' - \alpha_i p'$, which pins down a particular value for $\alpha_i = (q' - q)/(p' - p)$, which happens with zero probability, by the distribution of α_i . Also, the choice will never be empty: as they choose from a finite list of real numbers, one of them must be the highest. This means that, when integrating choice probabilities into market shares, we do not need to worry about multiplicity of choices.

Order the choices in $\widetilde{C}$ by quality, with $q_1 < q_2 < \dots < q_{\widetilde{J}}$. By construction, the same order applies to prices, $p_1 < p_2 < \dots < p_{\widetilde{J}}$, and to slopes. Finally, we can compute choice probabilities. Individual i chooses j if, and only if, $q_j - \alpha_i p_j \geq q_k - \alpha_i p_k$ for all pairs $(q_k, -p_k) \in \widetilde{C}$. But we can simplify the expression by only looking at neighbors, as in the Bresnahan vertical differentiation model, thanks to the decreasing slopes restriction.

Individual i prefers good j over $j + 1$ when $q_j - \alpha_i p_j \geq q_{j+1} - \alpha_i p_{j+1}$. Assume, by contradiction, that he prefers good $j + 2$ over j , so $q_j - \alpha_i p_j \leq q_{j+2} - \alpha_i p_{j+2}$. These two imply

$$\frac{q_{j+1} - q_j}{p_{j+1} - p_j} \leq \alpha_i \leq \frac{q_{j+2} - q_{j+1}}{p_{j+2} - p_{j+1}},$$

Which is impossible in our menu $\widetilde{C}$ by construction. As a result, if a consumer prefers j over $j + 1$, they will also prefer j over $j + 2$, and so on. Symmetrically, if he prefers j over $j - 1$, he will also

prefer j over all lower goods. Because of this, preferring j to both neighbors is equivalent to j being the global optimum. The probability that individual i chooses j is

$$\begin{aligned} P_{ij} &= P\left\{i \text{ prefers } j \text{ to } j+1 \text{ and to } j-1\right\} \\ &= P\left\{q_j - \alpha_i p_j \geq q_{j+1} - \alpha_i p_{j+1}, \quad q_j - \alpha_i p_j \geq q_{j-1} - \alpha_i p_{j-1}\right\} \\ &= P\left\{\frac{q_{j+1} - q_j}{p_{j+1} - p_j} \leq \alpha_i \leq \frac{q_j - q_{j-1}}{p_j - p_{j-1}}\right\}. \end{aligned}$$

For a given consumer that observes his α_i , this expression is either 0 or 1. In knife-edge cases where a consumer falls exactly at the margin, this would predict that they could choose more than one good. But this occurs with zero probability, so when we aggregate, each pair $(q_j, -p_j)$ in $\tilde{C}$ will have market share

$$s(q_j, -p_j) = \int_0^\infty \mathbb{I}\left[\frac{q_{j+1} - q_j}{p_{j+1} - p_j} \leq \alpha \leq \frac{q_j - q_{j-1}}{p_j - p_{j-1}}\right] dF(\alpha) = F\left(\frac{q_j - q_{j-1}}{p_j - p_{j-1}}\right) - F\left(\frac{q_{j+1} - q_j}{p_{j+1} - p_j}\right),$$

where F is the cdf of α_i . Define ties_j equal to the number of goods exactly tied in both quality and price with good j , including j . If j has no ties, $\text{ties}_j = 1$. To summarize, the market share of a good j is

$$s_j = \frac{s(q_j, -p_j)}{\text{ties}_j}, \text{ if } (q_j, -p_j) \in \tilde{C}, \quad \text{and 0 otherwise.}$$

For the worst good to have a well-defined market share, there is also the outside option of not buying anything, with $q_0 = p_0 = 0$. And for the best good, everyone with low enough α_i purchases it, so that its market share is $F(\dots)$, only with the first term.

ii. Bertrand-Nash price equilibrium

On the supply side, we assume a static multi product Bertrand-Nash game. Alan and Bianchi choose prices simultaneously, each taking the rival firm's prices as given. Product characteristics and marginal costs are taken as given, and each firm chooses the prices of all products it owns to maximize its total profit.

Alan owns $\mathcal{J}_A = \{a_1, a_2, a_3\}$ and Bianchi owns $\mathcal{J}_B = \{b_1, b_2\}$. Firm f 's profit in time t is

$$\pi_{ft} = \sum_{k \in \mathcal{J}_f} (p_{kt} - mc_{kt}) s_{kt}(\mathbf{p}_t).$$

Assuming an interior equilibrium and differentiable demand, the FOC for each $j \in \mathcal{J}_f$ is

$$s_{jt} + \sum_{k \in \mathcal{J}_f} (p_{kt} - mc_{kt}) \frac{\partial s_{kt}}{\partial p_{jt}} = 0.$$

To write these FOCs compactly, define the ownership matrix

$$\Omega_{kj} = \begin{cases} 1, & \text{if products } k \text{ and } j \text{ are owned by the same firm,} \\ 0, & \text{otherwise.} \end{cases}$$

Here,

$$\Omega = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

Thus Ω simply tells the model which cross price effects a firm internalizes. Alan internalizes the effects of changing one Alan price on all three Alan products, while Bianchi does the same for its two products. Neither firm internalizes the effect of its pricing decision on the rival firm's profit. We define

$$D_t = \left[\frac{\partial s_{kt}}{\partial p_{jt}} \right]_{k,j}$$

as the demand Jacobian. Then the five Bertrand first order conditions can be written as

$$\mathbf{s}_t + (\Omega \circ D_t)'(\mathbf{p}_t - \mathbf{mc}_t) = \mathbf{0},$$

where $\circ$ is the elementwise product. Defining

$$\Delta_t = -(\Omega \circ D_t)',$$

gives

$$\mathbf{s}_t - \Delta_t(\mathbf{p}_t - \mathbf{mc}_t) = \mathbf{0}.$$

If Δ_t is invertible,

$$\mathbf{p}_t - \mathbf{mc}_t = \Delta_t^{-1} \mathbf{s}_t.$$

iii. Product positioning and marginal costs

Again, we use the definition of quality as $q_j = \exp(\beta_w w_j + \beta_g g_j)$, with the parameter values from the question. Also plugging in the marginal cost functions from the question,

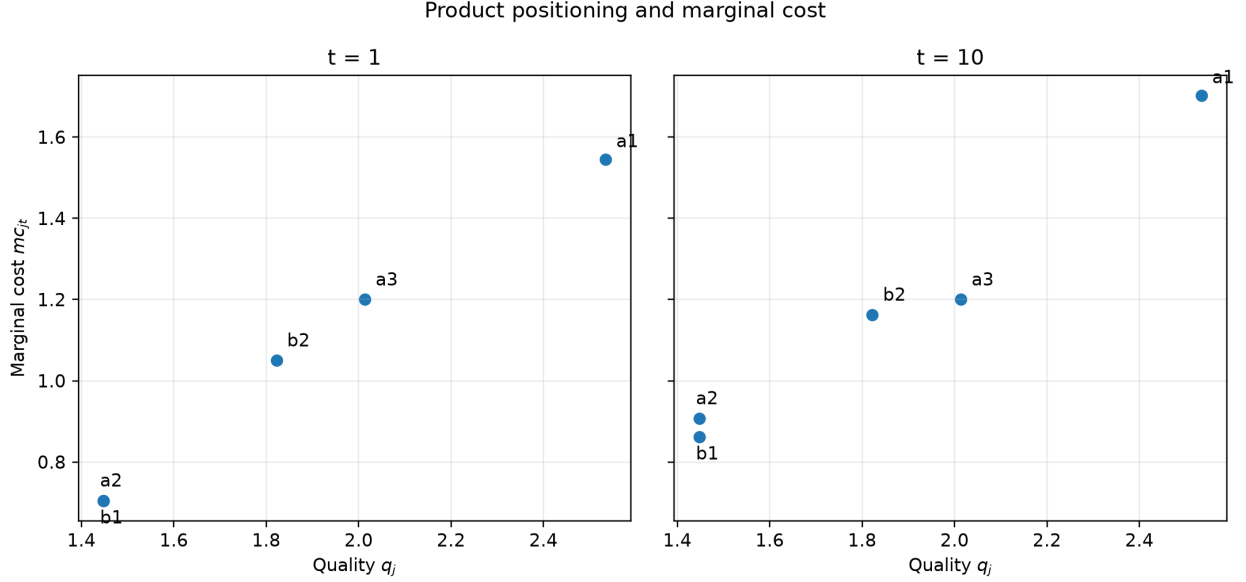


Figure 1: Product positioning and marginal cost at $t = 1$ and $t = 10$.

Note that for $t = 1$, a_2 and b_1 coincide as they have the same marginal cost. Also, we see that marginal cost rises over time for every product except a_3 , because $w_{a_3} = 1$ makes its carbon cost component equal to zero.

iv. Brief comment on market shares and competition

Product characteristics, and therefore quality, are fixed over time. The change in relative positioning therefore comes from marginal costs:

$$\Delta MC_j = \gamma_w(1 - w_j)(c_{10} - c_1) = 0.225(1 - w_j).$$

Hence,

$$(\Delta MC_{a_1}, \Delta MC_{a_2}, \Delta MC_{a_3}, \Delta MC_{b_1}, \Delta MC_{b_2}) = (0.1575, 0.2025, 0, 0.1575, 0.1125).$$

We expect a_3 to gain market share because its marginal cost does not increase. b_2 should also gain relative to products whose costs rise more, while a_2 , which has the largest cost increase, should lose.

The comparison between a_2 and b_1 is clear as they have identical quality (same value on x-axis), but by $t = 10$,

$$MC_{a_2,10} = 0.9075 > 0.8625 = MC_{b_1,10},$$

so b_1 gains a cost advantage over a_2 , and can try to bring prices down until a_2 is out of the market. Meanwhile, we see in the graph that the marginal costs of b_2 and a_3 become closer, such that b_2 should lose some competitive edge against a_3 , as they produce similar quality goods.

v. Bertrand NE for different μ

Calling the function that stacks all market shares $D(\mathbf{p}_t; \mu)$, we can substitute into the Bertrand-Nash FOC to have the equilibrium condition

$$D(\mathbf{p}_t; \mu) = \Delta_t(\mathbf{p}_t - \mathbf{mc}_t)$$

We can either solve for $\mathbf{p}_t$ and try to iterate until we find a fixed point, or just directly look for a root of the equation. But we run into a couple of problems related to ties if we try these.

First, if two firms produce goods with the exact same quality and marginal cost, the market share functions will be discontinuous in prices, and it will be difficult to find a solution because they'll be incentivised to undercut each other, and prices may jump around without converging. We can also not simply drop one of the goods, because that would change the jacobian and one of the firms would ignore the effects that their version of the good has. To deal with this, we impose the following assumption: if two different firms produce goods that are identical in both quality and marginal cost, they will then have directly Bertrand competition among them, such that both of them will price the good at $p_{jt} = mc_{jt}$ and equally split the market.

The second thing that causes problems is when different firms produce goods that are the same quality, but with differing marginal costs. Assume that happens for a good with $mc_A < mc_B$, for example. Firm A is able to undercut until firm B is out of the market, with $p = mc_B$. The problem is that, then, the best response of firm A is to price at its monopoly price. It may be the case that this new price will be above mc_B , incentivising firm B to rejoin the market. In this case, no Bertrand-Nash Equilibrium exists. The assumption we make is: if two firms produce the same quality good with $mc_A < mc_B$, then we remove firm B from the market. This can be interpreted as some entry cost that prevents firm B from reentering to compete again, even if prices are high enough.

To compute the equilibria, we start from a guess of which subset of goods are active, and an initial price vector with $p_j = q_j^2$, which guarantees that the slopes follow a decreasing order. Then, the code removes goods that are strictly dominated and deals with ties as we assumed above (it also checks if there are nondecreasing slopes). Then, we reduce the dimension of the problem to only non-dominated good that do not tie, in hopes that demand will be smooth. To search for the equilibrium, we use a least squares solver to find the price vector that minimizes the error in FOCs, looking only at the goods that remain in the market. As some of the prices are held fixed at some mc_j and some goods are eliminated from the start, this is a sort of partial equilibrium between the remaining ones.

Most of the time, this algorithm finds an equilibrium (very small deviations in FOCs of non-eliminated firms), but not always, in which case we still keep the smallest deviations the solver could find.

Below, as an example, are the market shares computed between times 1 and 10, for the case $\mu = 0$.

And the full datasets are in the `/output/` folder

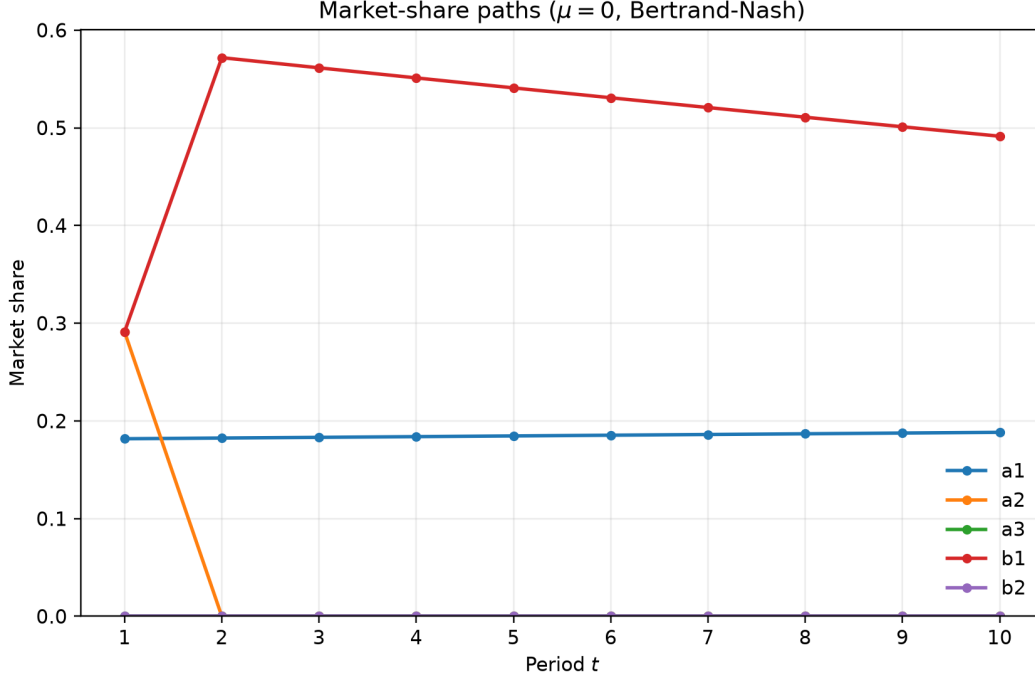


Figure 2: Market-share paths for the Bertrand-Nash equilibrium, $\mu = 0$.

Bertrand results: $\mu = -0.5$

Bertrand results: $\mu = 0$

Bertrand results: $\mu = 0.5$

vi. Coordination in a_2 and b_1 prices

Here Alan choose p_{a_1} and p_{a_3} to maximize π_A , Bianchi choose p_{b_2} to maximize π_B , and the pair (p_{a_2}, p_{b_1}) is chosen jointly to maximize total industry profit $\pi_A + \pi_B$ holding the other prices fixed. Thus the game has three price setting decision makers: Alan's noncoordinated products, Bianchi's noncoordinated product, and the coordinated (a_2, b_1) pair. We iterate these three best responses and verify that no decision maker has a profitable deviation. The agreement is valuable because $\delta_{a_2} = \delta_{b_1}$.

The two products have the same deterministic quality and, at $t = 1$, the same marginal cost, so they are very close substitutes. Under competition, cutting the price of one steals demand from the other. Coordinating their prices lets the firms take this effect into considerations and reduces the incentive to undercut each other.

Under collusion, it looks a lot more stable

And full datasets are

Collusion results: $\mu = -0.5$

Collusion results: $\mu = 0$

Collusion results: $\mu = 0.5$

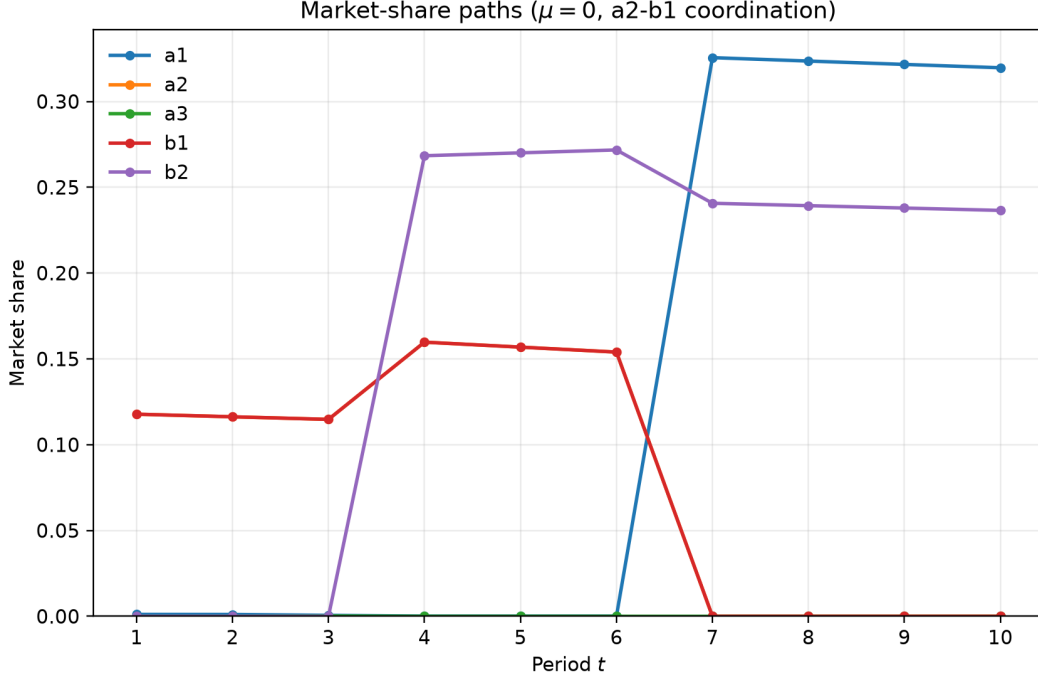


Figure 3: Market-share paths with $a2$ – $b1$ coordination, $\mu = 0$.

Part 2: Estimation

i. Estimator

Unlike Bresnahan, who estimates supply and demand parameters jointly, we estimate demand parameters first, then use them to back out marginal costs, then estimate the marginal cost parameters. Let $\theta_d = (\beta_w, \beta_g, \mu)$ denote the demand parameters we are looking to estimate. We can write our demand function from before as $D(\mathbf{p}; \theta_d)$, outputs predicted market share vectors. From the data, we observe shares s_{jt} for each good j at time t . We assume that market shares are observed with some prediction error $s_{jt} = D_j(\mathbf{p}; \theta_d) + \varepsilon_{jt}$, where $\varepsilon_{jt} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma^2)$. Under this assumption, maximizing the log-likelihood is equivalent to solving

$$\min_{\theta_d} \sum_t \left\| D(\mathbf{p}; \theta_d) - \mathbf{s}_t \right\|^2.$$

From this, we obtain $\hat{\theta}_d$. Then, we compute the analytical jacobian of the estimated demand function to build the Δ matrix for both the Bertrand and collusive ownership matrices. Here, we run into a problem with the inversion of Δ : as some goods have the same prices, it becomes singular. To circumvent this, we impose the constraint that, when two prices are equal, we assume they are in direct Bertrand competition with each other, hence we impose that their marginal costs will be equal to prices. Doing this, the remaining goods allow for an invertible matrix with which we solve for marginal costs.

One problem with this is that we cannot distinguish it to the case where two goods are prices exactly the same, but at their monopoly prices due to collusion. This can bias results towards assuming more competition than may exist. It also makes the Bertrand and Collusion assumptions generate the same results, as the two colluding goods are either priced the same, or one is out of

the market. We use this method to solve for each mc_{jt} . Finally, we run the linear regression

$$mc_{jt} = \gamma_w(1 - w_{jt})c_t + \gamma_g g_{jt} + u_{jt},$$

in order to estimate the supply parameters $\theta_s = (\gamma_w, \gamma_g)$. But, because marginal costs are estimated and not observed, the usual standard errors would be underestimated, as they do not take into account the uncertainty of the first stage.

ii. Results

Below are our estimation results. We chose to include several significant digits to show that they do not exactly match the ones in the data, although they are extremely close. The parameter values were not used as initial conditions, so they were actually found as an optimum.

Data	Scenario μ	$\hat{\beta}_w$	$\hat{\beta}_g$	$\hat{\mu}$	Share RMSE
Bertrand data	-0.5	-0.0999999981	0.1999999998	-0.5000000017	1.509667317e-10
Bertrand data	0	-0.0999999981	0.1999999998	-9.24703815e-10	2.256818339e-10
Bertrand data	0.5	-0.0999999981	0.1999999998	0.5000000004	3.848213675e-10
Collusion data	-0.5	-0.1000000096	0.2000000008	-0.5000000097	2.469157625e-09
Collusion data	0	-0.1000000096	0.2000000008	1.173668822e-09	1.728630892e-09
Collusion data	0.5	-0.1000000096	0.2000000008	0.5000000008	1.703749281e-16

The big downside is that, as we were only able to invert the Δ matrix to solve for marginal costs by imposing the assumption that, if $p_1 = p_2$, then $mc_1 = p_1 = p_2$, the different ownership matrices end up not making a difference. The second reason is that, in our simulations, a lot of the goods are out of the market, producing 0, and we also did not use those. Therefore, we cannot use our estimations to identify whether or not the firms were colluding in each case. Also, the supply parameter estimates are very far from the true values.

Data	Ownership	$\hat{\gamma}_w$	$\hat{\gamma}_g$	Recovered MCs	Cost RMSE
Bertrand data	Bertrand ownership	61.44021092	-4.526648296	150	35.90112601
Bertrand data	Collusive ownership	61.44021092	-4.526648296	150	35.90112601
Collusion data	Bertrand ownership	-0.5000346479	1.617698247	150	5.02359241
Collusion data	Collusive ownership	-0.5000346479	1.617698247	150	5.02359241

Part 3: Extra and optional

i. Households choosing multiple bikes

Let household h have $n_h \in \{1, \dots, 5\}$ members. In the spirit of Hendel (1999), we assume that each member can buy at most one bike, so the household chooses a bundle

$$\mathbf{x}_h = (x_{ha_1}, \dots, x_{hb_2}), \quad x_{hj} \in \mathbb{Z}_+, \quad \sum_j x_{hj} \leq n_h.$$

We model the household's utility as the sum of the utilities from all household members, with each one having a different α_i . We assume that each family member makes exactly the same choices as before, and they get aggregated into household purchases. Because of this, we have the same intervals as before: individual with $\alpha_i \in I_j$ will prefer to purchase bike I_j . Therefore, aggregating these choices, each household has demand

$$x_{hj} = \sum_{i=1}^{n_h} \mathbb{I}[\alpha_i \in I_j].$$

Integrating over the distribution of α_i , the total market demand is, in expectation,

$$x_j = \sum_{h=1}^H \sum_{i=1}^{n_h} P\{\alpha_i \in I_j\}.$$

But, because the problem has the same structure as before in the individual level, $P\{\alpha_i \in I_j\}$ is the same as our market shares from before, $D_j(\mathbf{p})$. If we only observe the aggregate purchases of each bike, adding the family structure with nothing extra does not meaningfully change the model. If we, for example, were to add some correlation in choices within choices by adding a common error term to all members of the same household, it becomes more interesting. To take advantage of the log normal assumption, assume that each individual i in household j , instead of just α_i , has $\alpha_i \times \beta_h$, with β_h also independently drawn from a log normal from each family, multiplying prices in the utility. Then, aggregate demand will be

$$x_j = \sum_{h=1}^H \sum_{i=1}^{n_h} P\{\alpha_i \beta_h \in I_j\}.$$

If we symbolically write I_j/β_h as the interval with both endpoints divided by β_h , we have $P\{\alpha_i \in I_j/\beta_h\}$. What is interesting about this is that, as the intervals are made up of terms like $(q_j - q_{j-1})/(p_j - p_{j-1})$, dividing everything by β_h is the same as multiplying all prices by β_h . It is as if each household observes distorted prices. In expected value,

$$\mathbb{E}_\alpha[x_j] = \sum_{h=1}^H \sum_{i=1}^{n_h} D_j(\beta_h \mathbf{p}) = \sum_{h=1}^H n_h D_j(\beta_h \mathbf{p}),$$

where we have integrated over individual heterogeneity, but still kept household heterogeneity. In fact, the expected (only over α) demand for good j from each household is $n_h D_j(\beta_h \mathbf{p})$. If we observe household size and the bundle each household chooses, we can estimate demand from those bundle choices, using the same functional form for D_j as we had before. As we still have to estimate the

variance of β_h (or of its log), for each guess of utility parameter values, we can pick the σ_β^2 that maximizes the likelihood that household demands we observe have β_h values that stem from a log normal with that parameter. This way, each set of utility parameters θ_d will map to one parameter for $\log \beta_h \sim \mathcal{N}(0, \sigma_\beta^2)$. So, for each guess of θ_d we can compute the expected demand of each family, and try to find the parameters that minimize its distance to the observed demand.